

Tutorial 6

Vector spaces

1. *This question asks you to check the axioms for a vector space in a specific example. You will need to have your notes open at the definition of a vector space (page 91).*

Let $\mathbb{R}^2 = \{(x_1, x_2) : x_1, x_2 \in \mathbb{R}\}$. We have seen that, with the usual definitions for addition and scalar multiplication of vectors, $\mathbb{R}^2$ is a vector space. In this question we use the usual scalar multiplication, but define a **different addition** to the usual one on $\mathbb{R}^2$ (we'll denote it by $\oplus$ to emphasize the difference). Your task will be to test whether $\mathbb{R}^2$ is a vector space with these two operations.

- Addition on $\mathbb{R}^2$ is defined as follows: For $\mathbf{x} = (x_1, x_2)$ and $\mathbf{y} = (y_1, y_2)$ in $\mathbb{R}^2$,

$$\mathbf{x} \oplus \mathbf{y} = (x_1 + y_1 + 1, x_2 + y_2 - 2).$$

- Scalar multiplication on $\mathbb{R}^2$ is defined as follows: For $\mathbf{x} = (x_1, x_2)$ in $\mathbb{R}^2$ and $\lambda \in \mathbb{R}$,

$$\lambda \mathbf{x} = (\lambda x_1, \lambda x_2).$$

- (a) Check that $\mathbf{x} \oplus \mathbf{y} \in \mathbb{R}^2$ and $\lambda \mathbf{x} \in \mathbb{R}^2$. (This is close to obvious; simply say why it is obvious.)

Solution: It is clear that if $\mathbf{x} = (x_1, x_2), \mathbf{y} = (y_1, y_2) \in \mathbb{R}^2$, then $\mathbf{x} \oplus \mathbf{y} = (x_1 + y_1 + 1, x_2 + y_2 - 2) \in \mathbb{R}^2$, and if $\lambda \in \mathbb{R}$, then $\lambda(x_1, x_2) = (\lambda x_1, \lambda x_2) \in \mathbb{R}^2$.

We now want to see whether $\mathbb{R}^2$ is a vector space over $\mathbb{R}$ when addition and scalar multiplication is defined as above. This will only be the case if *every one* of the eight axioms listed in Definition 2.1.4 for a vector space hold for $\mathbb{R}^2$ with the above addition and scalar multiplication.

- (b) Do the axioms 1 to 4 hold? If you claim an axiom holds, you must show that it holds (write out your check in full). If you claim that it does not hold, you must give an example to show that it does not hold.

[*Hint:* The zero vector and negatives may not be quite what you expect. Get your work for axioms 3 and 4 checked by a tutor.]

Solution: We check the first four axioms.

(1)

$$\mathbf{x} \oplus \mathbf{y} = (x_1 + y_1 + 1, x_2 + y_2 - 2) = (y_1 + x_1 + 1, y_2 + x_2 - 2) = \mathbf{y} \oplus \mathbf{x}.$$

(2)

$$\begin{aligned}(\mathbf{x} \oplus \mathbf{y}) \oplus \mathbf{z} &= (x_1 + y_1 + 1, x_2 + y_2 - 2) \oplus (z_1, z_2) \\&= ((x_1 + y_1 + 1) + z_1 + 1, (x_2 + y_2 - 2) + z_2 - 2) \\&= ((x_1 + (y_1 + z_1 + 1) + 1, x_2 + (y_2 + z_2 - 2) - 2) \\&= \mathbf{x} \oplus (\mathbf{y} \oplus \mathbf{z}).\end{aligned}$$

(3) For axiom 3, we need a vector, let's call it $\mathbf{z} = (z_1, z_2)$, such that for every $\mathbf{x} = (x_1, x_2) \in \mathbb{R}^2$, $(x_1, x_2) = \mathbf{x} = \mathbf{x} \oplus \mathbf{z} = (x_1 + z_1 + 1, x_2 + z_2 - 2)$. This means that we must have $z_1 = -1, z_2 = 2$. Therefore axiom 3 will be satisfied if we take as zero vector for $\mathbb{R}^2$ the vector $\mathbf{z} = (-1, 2)$.

(4) For axiom 4 we need, for every $\mathbf{x} = (x_1, x_2) \in \mathbb{R}^2$, a negative, let's say $\mathbf{y} = (y_1, y_2) \in \mathbb{R}^2$ such that $\mathbf{x} \oplus \mathbf{y} = \mathbf{z}$, where $\mathbf{z}$ is the zero vector found above. Check that $\mathbf{y} = (-x_1 - 2, -x_2 + 4)$ satisfies this equation. Clearly $\mathbf{y}$ is in $\mathbb{R}^2$.

This shows that the axiom 1 to 4 are satisfied.

- (c) Do the axioms 5 to 8 all hold? Again, give proofs or counter-examples to support your claims.

Solution:

(5) For $\lambda \in \mathbb{R}$, $\mathbf{x} = (x_1, x_2), \mathbf{y} = (y_1, y_2) \in \mathbb{R}^2$ we have $\lambda(\mathbf{x} \oplus \mathbf{y}) = (\lambda x_1 + \lambda y_1 + \lambda, \lambda x_2 + \lambda y_2 - 2\lambda)$ while $\lambda\mathbf{x} \oplus \lambda\mathbf{y} = (\lambda x_1 + \lambda y_1 + 1, \lambda x_2 + \lambda y_2 - 2)$ and these are in general **not** equal (take $\lambda = 2$ for example), so the axiom 5 does not hold.

(6) For $\lambda, \mu \in \mathbb{R}$, $\mathbf{x} \in \mathbb{R}^2$ we have $(\lambda + \mu)\mathbf{x} = ((\lambda + \mu)x_1, (\lambda + \mu)x_2)$ while $\lambda\mathbf{x} \oplus \mu\mathbf{x} = (\lambda x_1 + \mu x_1 + 1, \lambda x_2 + \mu x_2 - 2)$, and these two are in general **not** equal. (Take, for example $\lambda = \mu = 1$). So axiom 6 does not hold.

(7) For λ, μ and $\mathbf{x}$ as above, check that $(\lambda\mu)\mathbf{x} = \lambda(\mu\mathbf{x})$, so axiom 7 holds.

(8) For $\mathbf{x} \in \mathbb{R}^2$ $1\mathbf{x} = \mathbf{x}$, so axiom 8 holds.

- (d) Is $\mathbb{R}^2$ a vector space with the addition and scalar multiplication as defined above?

Solution: $\mathbb{R}^2$ is **not** a vector space with the addition $\oplus$ and usual scalar multiplication, since two of the axioms fail. (You can already say this as soon as you find *one* axiom that fails.)

2. Let V be a vector space.

- (a) Let $\mathbf{u} \in V$. Prove that $\mathbf{u} + \mathbf{u} = 2\mathbf{u}$. This is not one of the axioms of a vector space, so it *does* need proof. Be careful to use *only* the definition of a vector space in your proof; it is a good discipline to say at each step in your proof which axiom of the definition you are using. (*Hint: $\mathbf{u} = 1\mathbf{u}$.*)

Solution:

$$\begin{aligned}\mathbf{u} + \mathbf{u} &= 1\mathbf{u} + 1\mathbf{u} && \text{(using axiom 8 of vector spaces)} \\ &= (1 + 1)\mathbf{u} && \text{(using axiom 6 of vector spaces)} \\ &= 2\mathbf{u} && \text{(using real number addition!)}\end{aligned}$$

- (b) Prove Theorem 2.1.12(c), which says that in any vector space V , $\lambda\mathbf{0} = \mathbf{0}$ for any scalar λ . Again, be careful to use only the definition of a vector space in your proof. (*Hint: $\mathbf{0} = \mathbf{0} + \mathbf{0}$.*)

Solution: Hint: Use $\lambda(\mathbf{0} + \mathbf{0}) = \lambda\mathbf{0}$ and axiom 5.

- (c) Prove that in any vector space V , $\lambda\mathbf{u} = \mathbf{0}$ if and only if $\lambda = 0$ or $\mathbf{u} = \mathbf{0}$ (See Theorem 2.1.12(e)).

Solution: Theorem 2.1.12 (b) and (c) give the one implication. For the other implication we can use a proof by contradiction: Suppose $\lambda\mathbf{u} = \mathbf{0}$ and $\lambda \neq 0$ and $\mathbf{u} \neq \mathbf{0}$. Then

$$\mathbf{0} = \lambda^{-1}\mathbf{0} = \lambda^{-1}(\lambda\mathbf{u}) = (\lambda^{-1}\lambda)\mathbf{u} = 1\mathbf{u} = \mathbf{u},$$

a contradiction. Make sure that you can give a reason for every single one of the equalities listed here.

- (d) Do the two statements you proved in (a) and (b) hold in the set $\mathbb{R}^2$ of Question 1, with the addition and scalar multiplication defined there? And the statement $\lambda\mathbf{u} = \mathbf{0} \Rightarrow \lambda = 0$ or $\mathbf{u} = \mathbf{0}$? Give a reason for your answer.

Solution: The property in (a) does not hold (take $\mathbf{u} = (0, 0)$ for example). The other two statements also do not hold; try to find your own counter examples.

3. It follows from the definition of a vector space that every vector space V comes equipped with two operations, that of **addition** and **scalar multiplication**.

- (a) Is it possible to use these two operations and the axioms of a vector space to define an operation of **subtraction** on V ? If so, how would you do it? (Your definition should work in any vector space, not just in a specific one.)

Solution: We may define subtraction in V by putting for $\mathbf{u}, \mathbf{v} \in V$,

$$\mathbf{u} - \mathbf{v} = \mathbf{u} + (-\mathbf{v}) = \mathbf{u} + (-1)\mathbf{v}.$$

- (b) Is the operation of subtraction you defined commutative?

Solution: The operation is in general not commutative.

Here is a possible counter-example: Take $V = \mathbb{R}^2$, with its usual addition and scalar multiplication. Then for instance

$$\begin{pmatrix} 1 \\ 2 \end{pmatrix} - \begin{pmatrix} 3 \\ 4 \end{pmatrix} = \begin{pmatrix} -2 \\ -2 \end{pmatrix}$$

while

$$\begin{pmatrix} 3 \\ 4 \end{pmatrix} - \begin{pmatrix} 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 2 \\ 2 \end{pmatrix}.$$

- (c) Is the operation of subtraction you defined associative?

Solution: The operation is in general not associative.

Here is a possible counter-example: Take $V = \mathbb{R}^2$, with its usual addition and scalar multiplication. Then for instance

$$\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix} - \begin{pmatrix} 1 \\ 1 \end{pmatrix} \right) - \begin{pmatrix} 2 \\ 2 \end{pmatrix} = \begin{pmatrix} -3 \\ -3 \end{pmatrix}$$

while

$$\begin{pmatrix} 0 \\ 0 \end{pmatrix} - \left(\begin{pmatrix} 1 \\ 1 \end{pmatrix} - \begin{pmatrix} 2 \\ 2 \end{pmatrix} \right) = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

- (d) Prove that the additive identity of V is also a “subtractive identity”.

Solution: The additive identity of V is the zero vector $\mathbf{0}$. Now it holds that for any $\mathbf{v} \in V$,

$$\mathbf{v} - \mathbf{0} = \mathbf{v} + (-\mathbf{0}) = \mathbf{v} + \mathbf{0} = \mathbf{v}.$$

(It is not hard to show that in any vector space, $-\mathbf{0} = \mathbf{0}$.)